pyladcp · ladcp-qa CLI guide

Process a whole cruise, step by step. Run the steps in order, in one terminal.

STEP 0 · SET THE CRUISE ROOT (FIRST!)

Every command uses $B = the cruise folder that holds raw_ladcp/, raw_CTD/ and sADCP/. Set it once:

```
cd /path/to/cruise/MORIA   # folder with raw_*
B="$PWD"                    # = where you are now
echo "$B"                   # MUST print that path
```

If echo "$B" prints nothing, STOP and set it again. An empty $B turns --ladcp "$B/raw_ladcp" into "/raw_ladcp" → "indexed 0 casts from 0 files". A new terminal forgets $B — re-set it.

YOUR FOLDER MUST LOOK LIKE THIS

```
$B/
├─ raw_ladcp/
│   ├─ MASTER/  *.000   down-looker
│   └─ SLAVE/   *.000   up-looker
├─ raw_CTD/   *.hex *.hdr *.bl *.XMLCON
│             (one set per station)
└─ sADCP/     ship-ADCP (optional)
    └─ sadcp_150/DATA/  *.STA
```

Raw LADCP files carry NO station number — the CTD .hex (station + GPS UTC) is the anchor that tells the index which raw file is which cast.

raw_ladcp/ may instead be a FLAT tree (no MASTER/SLAVE) — heads split by the PD0 facing bit. Beam-coord PD0s are rotated to earth on read.

NEW / NON-MORIA CRUISE

--cruise <NAME>: any name works. An unknown cruise gets shared operator defaults (dz 8, bin-1 mask, tilt 22/4, RDI-firmware bottom track), stamped <NAME> on the exports.

SADCP clock not synced to GPS (wrong dates in the STA files)? Recover the offset automatically:

```
ladcp-qa ... --sadcp-timeoff auto \
    --sadcp-nav "$B/Navigation"
```

slides the STA GPS against an independent nav track (SADO export or any time/lat/lon CSV). Shallow shelf cast leaking a bad near-bottom cell? raise the reject margin: --dzbelow 24.

STEP 1 · BUILD THE INDEX (ONCE)

Note: --root comes BEFORE the word build.

```
ladcp-index --root "$B" build \
    --ladcp "$B/raw_ladcp" \
    --ctd   "$B/raw_CTD"
```

Expect “indexed ~40 casts from ~100 files”.
If it says 0 casts / 0 files → $B is unset (Step 0).

STEP 2 · CHECK THE PAIRING

```
ladcp-index --root "$B" show
```

Lists each station with its master + slave file.
Confirm every cast got both before the long run.

STEP 3 · PROCESS THE WHOLE CRUISE

The full constrained inverse is the default solver.

```
ladcp-qa --all-stations \
    --index "$B/.ladcp_archive.json" \
    --root "$B" --cruise <NAME> \
    --from-hex --ctd-cache "$B/ctd_from_hex" \
    --sadcp "$B/sADCP/sadcp_150/DATA" \
    -o "$B/qa_out"
```

A live progress bar shows X/N stations. Full detail + any failures → $B/qa_out/ladcp-qa.log. One bad cast is logged and skipped, not fatal.

Then open qa_out/stations/<st>/<st>_report.pdf
→ scorecard (OK / WARN / FAIL) + the figures.

STEP 4 · COMPARE VS A LEGACY REFERENCE

Have LDEO *.mat from a prior processing? Pair by cast time (no filename assumptions) and score:

```
ladcp-compare --ours "$B/qa_out" \
    --legacy "$B/LADCP_processed" \
    -o "$B/qa_out/legacy_compare"
```

→ comparison.csv + comparison_report.pdf (u/v overlays with ±1σ bands). unpaired.txt lists any station matched on only one side — never dropped.

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Reference — what you get, every flag, and the things that trip people up.

WHAT YOU GET (UNDER $B/QA_OUT/)

```
qa_out/
├ stations/<MORIA-xx>/
│   ├ <st>_report.pdf    ⋆ scorecard+figs
│   ├ <st>.lad  <st>.bot velocity / b-track
│   ├ <st>.nc   <st>.xlsx NetCDF / Excel
│   ├ <st>_qa.txt/.json  QA metrics
│   └ figures/*.png
├ exports/               cruise-level
│   ├ MORIA_ladcp.xlsx / .nc
│   ├ MORIA_ladcp_odv.txt  (Ocean Data View)
│   └ MORIA_summary.csv
└ ladcp-qa.log           run log
```

exports/ is written by --all-stations (or add --cruise-export to a named list of stations).

COMMON VARIATIONS

One station, full solve + ship-ADCP:

```
ladcp-qa 80 --index "$B/.ladcp_archive.json" \
   --root "$B" --from-hex \
   --sadcp "$B/sADCP/sadcp_150/DATA" \
   -o "$B/qa_out"
```

A few stations + a cruise workbook:

```
ladcp-qa 79 80 82 ...  --cruise-export
```

No Excel? skip it (or install the extra):

```
ladcp-qa ... --formats odv,nc,csv
pip install pyladcp[export]   # enables .xlsx
```

Stream detail instead of the bar: add -v

CONSTRAINT WEIGHTS — PHYSICAL MEANING

The inverse blends four information sources. Each weight scales how strongly that source pulls the solution relative to the LADCP water-track data: 1 = legacy-standard balance · 0 = off · >1 = trust more.

`--botfac W`	bottom track (default 1). Velocity over ground from the seabed echo: anchors the absolute (barotropic) level near the bottom.
`--sadcpfac W`	ship ADCP (default 3). The hull ADCP's absolute currents: pins the upper-ocean bins.
`--barofac W`	GPS navigation (default 1). Ship drift over the cast: one row pinning the depth-mean (barotropic) velocity.
`--smoofac W`	vertical smoothing (default 0). Curvature penalty on the profile; at 0 only ill-constrained bins are smoothed (golden).

Where each constraint acts (and how hard) is drawn per cast: figures/<st>_weights.png (legacy Fig 12).

LADCP-INDEX (--ROOT BEFORE BUILD/SHOW)

`--root D`	cruise root paths are relative to
`build`	scan + cache station→file map
`  --ladcp D`	archive dir (has MASTER/ + SLAVE/)
`  --ctd D`	CTD .hex/.hdr dir (the anchor)
`  --rescan`	force full re-decode (ignore cache)
`show`	print the resolved casts

LADCP-QA (ORDER-FREE FLAGS)

`--all-stations`	every cast in the index + exports/
`--cruise NAME`	preset; unknown → operator defaults
`--cruise-export`	exports/ for a named station list
`--index F`	archive index → files per station
`--solver S`	inverse (default, full) | shear
`--sadcp D`	ship-ADCP source (inverse constraint)
`--sadcp-source X`	vmdas (default) | codas
`--sadcp-timeoff T`	STA clock fix [s] or 'auto'
`--sadcp-nav P`	nav track for --sadcp-timeoff auto
`--botfac W`	bottom-track weight (see page-left)
`--sadcpfac W`	ship-ADCP weight (default 3)
`--barofac W`	GPS-barotropic weight (default 1)
`--smoofac W`	smoothing weight (default 0)
`--dzbelow M`	near-seabed reject margin [m]
`--down-only`	solve from the down-looker alone
`--nearfield-dn-bins`	device-band mask override / 'none'
`--from-hex`	build CTD from raw .hex if no .cnv
`--ctd-cache D`	reuse dir for --from-hex
`--formats L`	xlsx,odv,nc,csv (default: all)
`-o D`	output folder (default qa_out)
`-v / --verbose`	stream detail instead of the bar
`--root D`	base for cleaned CTD/ + fallback find

GOTCHAS

- Set $B first and check echo "$B" (Step 0).
- ladcp-index: --root goes BEFORE build / show.
- Excel needs openpyxl (pip install pyladcp[export]); else .xlsx is skipped, the rest still write.
- --from-hex needs CTD_project; a clean .cnv wins.
- Heads pair by time, so fragmented / offset master-slave deployment files still match.
- battery is an uncalibrated estimate (0.33·xmv): it can WARN but never FAILs a cast.
- nearfield_errvel_ratio WARN >1.7 = a rigid target hung below the package (e.g. a corer); mask its bins per cruise (MORIA does 03–28 automatically).
- --down-only is a cross-check — don't deliver it where the down-looker is contaminated.
- Headless server? prefix MPLBACKEND=Agg.

LADCP-COMPARE (VS LEGACY LDEO .MAT)

`--ours D`	ladcp-qa out dir (stations/*/.nc)
`--legacy D`	legacy processed dir (dr *.mat)
`--alt-dir D`	sub a rerun in for --alt-stations
`--sadcp D`	overlay ship-ADCP on each cast
`-o D`	report dir → .csv + report .pdf